

Marcella J. Cross

Postdoctoral Research Associate ◊ mcross9@wisc.edu

Education:

University of British Columbia, PhD Student, Michaletz Lab	2022-2026
Cornell University, Ithaca, New York, Biological Sciences, Magna Cum Laude	2018-2022

Experience:

Postdoctoral Research Associate, Trowbridge Lab , University of Wisconsin-Madison, Department of Forest and Wildlife Ecology	2026-Present
Teaching Assistant, Biology 306, Advanced Ecology Professor: Sean Michaletz	2022-2025
Teaching Assistant, Biology 404, Ecological Methodology and Statistics Professor: Diane Srivastava	Spring 24-26
Instructor, Plant Functional Traits Course. Field research group instructor in the Drakensburg Mountains of South Africa. https://plantfunctionaltraitscourses.w.uib.no/	Winter 2023
Student Researcher, Rose Lab, Cornell University Phenotypic and biosynthetic analysis of tomato cuticle development	2021-2022
Crew Lead, Battles Lab, UC Berkeley, Forest Drought Mortality Field Work Field work with tree drought mortality patterns in the Sierra Nevada Mountains under Dr. John Battles and Dr. Jodi Axelson at Berkeley.	Summer 2021
Intern, Preston Lab, Plant Pathology, University of Oxford Work with <i>Pseudomonas syringae</i> infection of <i>Phaseolus vulgaris</i> . Skills include apoplast extraction, phaseolotoxin production, H ₂ O ₂ tolerance, bacterial mobility, etc.	Summer 2019
Student Researcher, Bauerle Lab, Soil Sciences - Cornell University Extracted DNA and ran PCR and gels from soil samples.	Spring 2019
Intern, Trillas Gay Lab, Plant Physiology, University of Barcelona Isolated and analyzed <i>Verticillium dahliae</i> in Cotton and Olive plants.	Spring 2018
Intern, Dr. Karim Ammar, International Maize and Wheat Improvement Center Ciudad Obregón, Mexico, emasculated and analyzed new rust-resistant wheat strains.	Winter 2018
Intern, Advanced Nanotechnologies, Barcelona, Spain Developed a transparent heater technology for the automobile industry.	Fall 2017
Research Assistant, Dr. Andrew Whiteley, University of Montana Genetic rescue project sampling and tagging native trout populations while camping in Montana wilderness at field sites.	Summer 2017
Intern, Kelly Lab, Fish and Wildlife, Virginia Tech Analyzed camera trapping data to count jaguars of Belize.	Summer 2016
Intern, O'Rourke Lab, Horticulture, Virginia Tech Studied native wildflower establishment in Virginia.	Summer 2016
Intern, Ecology Project International, Missoula, MT Compared mycorrhizal colonization of native and invasive plants in lab and field trials.	Summer 2015

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Presentations and Publications:

- Cross, M. (2026). Temperature responses of plant gas exchange from leaves to ecosystems. University of British Columbia. Thesis retrieved from <https://open.library.ubc.ca/collections/ubctheses/24/items/1.0452539> 2026
- Cross, M., and Michaletz, S.T. (2025). Excision increases branch water Potential but reduces leaf gas exchange. Tree Physiology, 2026, <https://doi.org/10.1093/treephys/tpag035> 2026
- Halbritter, A.H., Vandvik, V., ... Cross, M. et al. Plant traits and associated ecological data from Afromontane grasslands of Maloti-Drakensberg, South Africa. Sci Data 12, 1778 (2025). <https://doi.org/10.1038/s41597-025-06045-x> 2025
- Philippe, G., Sørensen, I., ... Cross, M. et al. Spatiotemporal variation in cutin polymerization and remodeling mediated by GDSL-hydrolase enzymes during tomato fruit development. BioRxiv pre-print: <https://doi.org/10.1101/2025.01.09.632122> 2025
- Decoupling of photosynthesis and evapotranspiration in heatwaves. American Geophysical Union. New Orleans, LA. Poster presentation Dec. 2025
- Decoupling of photosynthesis and evapotranspiration in heatwaves. Ecological Society of America. Baltimore, MD. Oral presentation. August 2025
- Plant optimal temperature vs. elevation: controlling for VPD. Ecological Society of America. Longbeach, CA. Oral presentation. August 2024
- Cross, M., and Bauerle, T. (2024). Detecting Plant Health via Electrical Stress Response. Authorea pre-print. DOI: 10.22541/au.170670050.05605607/v1 July 2024
- Excision effects on photosynthetic measurements in drought stressed and resinous plants. EcoEvo Conference. Oral Presentation. October 2023
- Detecting plant health via electrical stress response. Dr. Michele Holbrook's lab Lab meeting presentation. Harvard University October 2021
- Detecting plant health via electrical stress response. Ecological Society of America. Longbeach, CA. Oral presentation. August 2021
- Presentation Link: <https://youtu.be/f8Sp3cBcoj0>
- Poster: Cross, M. and Michaletz, S. 2023. Drivers of terrestrial net primary productivity. STAIR grant poster session, UBC. April 2023
- Poster: Cross, M. and Michaletz, S. 2022. Excision effects on photosynthetic measurements in drought-stressed and resinous plants. EcoEvo Conference. October 2022

Public Service:

- Mentor, Biology Undergraduate Diversity in Research. <https://blogs.ubc.ca/budr/> 2023-2026
- Peer-Mentor, UBC Botany Graduate Program 2023-2026
- Volunteer Mentor, [Rural eMentoring Program](#). Mentoring high school students. 2024-2026

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Volunteer, Fall Creek Elementary School 5th grade Math Program	Spring 2022
Volunteered through Cornell EDUC 3110 class 5 hours per week.	
Member, Scientista, Cornell University	2020-2022
Volunteer, Friends of Farmworkers, Cornell University	2018-2022
English as a second language tutoring for local farmworkers.	
Volunteer, YMCA Children's Program, Barcelona, Spain	Winter 2017
Volunteer, Fundació Pare Manel, Children's Program, Barcelona, Spain	Winter 2017

Publications and Awards:

President's Academic Excellence Initiative PhD Award	2022-2026
UBC Faculty of Science PhD Tuition Award	2022-2026
UBC International Tuition Award	2022-2026
Polk County Farm Bureau Scholarship Award	2018-2025
Oregon Association of Nurseries Scholarship Award	2021-2026
UBC Graduate Student Travel Award	Winter 2024
Simons Foundation Doctoral Scholarship	2022
Dean's List, Cornell University, 3.5 GPA or Above	2018-2022

References:

- **Amy Trowbridge** (Postdoc supervisor)
Assistant Professor, University of Wisconsin-Madison
amtrowbridge@wisc.edu (preferred method of contact)
(608) 263-4520
- **Sean Michaletz** (PhD supervisor)
Professor of Botany, University of British Columbia
sean.michaletz@ubc.ca (preferred method of contact)
(604) 822-2141
- **Diane Srivastava** (TA supervisor)
Professor of Zoology, University of British Columbia
srivast@zoology.ubc.ca (preferred method of contact)
- **Amy Angert** (Committee member)
Professor of Botany, University of British Columbia
amy.angert@botany.ubc.ca (preferred method of contact)
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